

TEK4090 – Introduction to Modern Control

Assignment 2 v4.0

Due: October 23 2024, 13:15

1. (10 points) Consider the continuous-time system,

$$\ddot{x} = u \tag{1}$$

$$J(t_0) = \frac{1}{2}x^2(0) + r \int_0^2 u^2 dt. \tag{2}$$

Hint: read §6.3 in Lewis, Syrmos, & Vraibie for some relevant example problems

- (a) Write the Hamilton-Jacobi-Bellman (HJB equation), eliminating $u(t)$

Ans: We begin by writing the hamiltonion equation:

$$H(x, u, \lambda, t) = ru^2 + \lambda^T f(x, u, t)$$

Where $\lambda \in \mathbb{R}^2$. We define f as a vector in $\mathbb{R}^2$ as well where

$$f = \begin{bmatrix} x_2 \\ u \end{bmatrix}$$

if we take $\dot{x} = x_2$ and $\dot{x}_2 = u$. We not get that the Hamiltonion is

$$H = ru^2 + \lambda_1 x_2 + \lambda_2 u$$

Our goal is to find the Hamilton Jacobi equation,

$$-J_t^* = \min_u (ru^2 + \lambda_1 x_2 + \lambda_2 u)$$

which we find by optimizing H for u and substituting the optimal u in H. First note that $\frac{\partial^2 H}{\partial u^2} = 2r \geq 0$, which is a condition for global convexity, assuming $r \geq 0$. This means that the optimal u^* we find is the global minimum of the Hamiltonion.

$$\begin{aligned} \frac{\partial H}{\partial u} &= 0 \\ &= 2ru + \lambda_2 \\ u^* &= -\frac{\lambda_2}{2r} \end{aligned}$$

We now substitute optimal control into h

$$\begin{aligned} H^* &= r \left(-\frac{\lambda_2}{2r} \right)^2 + \lambda_1 x_2 - \frac{\lambda_2^2}{2r} \\ &= \frac{\lambda_2}{4r} + \lambda_1 x_2 - \frac{\lambda_2^2}{2r} \\ &= \frac{-\lambda_2^2}{4r} + \lambda_1 x_2 \end{aligned}$$

Substituting this into J^* we get that

$$-J_t^* = \min_u (ru^2 + \lambda_1 x_2 + \lambda_2 u) = \frac{-\lambda_2^2}{4r} + \lambda_1 x_2$$

(b) Assume that the optimal cost-to-go function $J^*(t)$ is given by,

$$J^*(t) = \frac{1}{2}s_1(t)x^2(t) + s_2(t)x(t)\dot{x}(t) + \frac{1}{2}s_3(t)\dot{x}^2(t). \quad (3)$$

Use the HJB equation to derive coupled differential equations for these $s_i(t)$'s, and find boundary conditions for these equations.

Ans: We find that

- $\lambda_1 = \frac{\partial J^*}{\partial x} = s_1 x + s_2 \dot{x}$
- $\lambda_2 = \frac{\partial J^*}{\partial \dot{x}} = s_2 x + s_3 \dot{x}$
- $\frac{\partial J^*}{\partial t} = \frac{1}{2}s_1 \dot{x}^2 + s_2 x \ddot{x} + \frac{1}{2}s_3 \ddot{x}^2$

If we plug this into the HJB equation we found from a), we get

$$\begin{aligned} - \left[\frac{\partial J^*}{\partial t} = \frac{1}{2}\dot{s}_1 x^2 + \dot{s}_2 s_2 x \dot{x} + \frac{1}{2}\dot{s}_3 \dot{x}^2 \right] &= \frac{-1}{4r} (s_2 x + s_3 \dot{x})^2 + (s_1 x + s_2 \dot{x}) \dot{x} \\ &= \frac{-1}{4r} (s_2^2 x^2 + 2s_2 s_3 x \dot{x} + s_3^2 \dot{x}^2) + s_1 x \dot{x} + s_2 \dot{x}^2 \\ &= \frac{-s_2^2 x^2}{4r} + x \dot{x} \left(s_1 - \frac{s_2 s_3}{2r} \right) + \dot{x}^2 \left(s_2 - \frac{s_3^2}{4r} \right) \end{aligned}$$

Now we can match the coefficients with the original J^* and get

$$-\frac{1}{2}\dot{s}_1 x^2 - \dot{s}_2 x \dot{x} - \frac{1}{2}\dot{s}_3 \dot{x}^2 = \frac{-s_2^2 x^2}{4r} + x \dot{x} \left(s_1 - \frac{s_2 s_3}{2r} \right) + \dot{x}^2 \left(s_2 - \frac{s_3^2}{4r} \right)$$

- $-\frac{1}{2}\dot{s}_1 = -\frac{s_2^2}{4r}$
- $-\dot{s}_2 = s_1 - \frac{s_2 s_3}{2r}$
- $-\frac{1}{2}\dot{s}_3 = s_2 - \frac{s_3^2}{4r}$

Which makes

- $\dot{s}_1 = \frac{s_2^2}{2r}$
- $\dot{s}_2 = \frac{s_2 s_3}{2r} - s_1$
- $\dot{s}_3 = \frac{s_3^2}{2r} - 2s_2$

Now for the initial conditions. At $t = 2 = T$, cost to go is $J^*(T) = \frac{1}{2}x(T)^2$. From $J^* = \frac{1}{2}s_1(T)x^2 + s_2(T)xx + s_3(T)\dot{x}^2$, we draw that

- $s_1(T) = 1$
- $s_2(T) = 0$
- $s_3(T) = 0$

- (c) Write the linear state feedback optimal control policy in terms of the $s_i(t)$'s.
2. (10 points) Pick a system from a paper that you have found on a topic that you are interested in, potentially the topic you will choose for your final project.
- (a) What is/are the paper(s) you are drawing from?
 - (b) What are the dynamics of the system in question? What are the inputs and outputs?
 - (c) What is the system trying to achieve?
 - (d) What are the challenges for the system?
 - (e) What control tools can help the system overcome the challenges in achieving its goals?
3. (20 points) Suppose that $f : \mathbb{R} \mapsto \mathbb{R}$ is convex, and $a < b$ are in the domain of f .
- (a) Show that

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) \quad (4)$$

for all $x \in [a, b]$

Ans: A criteria for a function to be convex is that

$$f(x) \leq \lambda f(a) + (1 - \lambda) f(b)$$

$\forall x \in \mathbb{R}$

If we set $x = \lambda a + (1 - \lambda) b$ where $\lambda = \frac{b-x}{b-a}$, then

$$\begin{aligned} 1 - \lambda &= 1 - \frac{b-x}{b-a} \\ &= \frac{(b-a) - (b-x)}{b-a} \\ &= \frac{x-a}{b-a} \end{aligned}$$

And we get that

$$f(x) = f(\lambda a + (1 - \lambda) b) \leq \lambda f(a) + (1 - \lambda) f(b)$$

Which is the criteria for a function to be convex, and it is proved.

(b) Show that

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x} \quad (5)$$

for all $x \in (a, b)$. Draw a sketch that illustrates this inequality

Ans: If we pick up our old calculus text book, we find that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ iff. $a \leq b \forall c \in \mathbb{R}$

$$\frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(c)}{b - c}$$

If we assume that f is convex, then the line between the points $(a, f(a))$ and $(b, f(b))$ will never be under the point $(c, f(c))$.

1. Lets now denote the slope of the line $d((a, f(a)), (b, f(b)))$ as k . In this case, we use the $\frac{\text{rise}}{\text{run}}$ formula to get that $\frac{f(b)-f(a)}{b-a} = k$
2. The line $d((c, f(c)), (a, f(a)))$ as the line $\frac{f(c)-f(a)}{c-a} = k_1$

3. And finally, the line $d((b, f(b)), (c, f(c)))$ as the line $\frac{f(b)-f(c)}{b-c} = k_2$

If f is truly convex, then $k_1 \leq k \leq k_2$. Therefore,

$$\frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(c)}{b - c}$$

Now we can just replace c with x , because this inequality is valid $\forall x \in (a, b)$

(c) Suppose that f is differentiable. Use the result in (b) to show that

$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b). \quad (6)$$

Ans: From b we have that

$$\frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(c)}{b - c}$$

We know from a definition of derivatives in our calculus book, that

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

. Then since $x - b \leq 0$ we have that

$$f'(b) = \lim_{x \rightarrow b} \frac{f(b) - f(x)}{b - x}$$

From this we get what we wanted to prove, if we allow x to approach both a and b respectively

(d) Suppose f is twice differentiable. Use the result in (c) to show that $f''(a) \geq 0$ and $f''(b) \geq 0$.

Ans: Assume that f is convex, then, $\forall c \in (a, b)$ we have $\frac{f(c)-f(a)}{c-a} \leq \frac{f(b)-f(c)}{b-c}$. From the mean value theorem, $\exists c_1, c_2 \in (a, c), (c, b)$ s.t. $\frac{f(c)-f(a)}{c-a} = f'(c_1)$ and $\frac{f(b)-f(c)}{b-c} = f'(c_2)$

Since we have that $a < c_1 < c < c_2 < b$ and f is convex, $f'(c_1) \leq f'(c_2)$, and we get that $f'(x) \geq 0 \forall x \in (a, b)$

Ans:

(a)

4. (20 points) (LQR for Buildings) Consider a multi-zone building as in Fig. 2.

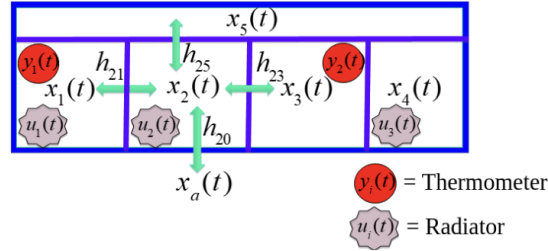


Fig. 2: Building model schematic

- (a) Write down the dynamical equations of the building in state-space form, including a term for disturbances $d(t)$, and for the ambient temperature $x_a(t)$, assuming the parameters are

$$\frac{h_{ij}}{C_i} = \begin{cases} 10 \text{ W/}^\circ\text{K} & i \text{ is adjacent to } j \\ 0 & \text{else} \end{cases} \quad (7)$$

$$C_i = 1.08 \times 10^6 \text{ J/}^\circ\text{K} \quad (8)$$

- (b) Consider an LQR cost function of the form

$$\int_0^T [(x - \bar{x})^T Q (x - \bar{x}) + u^T R u] dt, \quad (9)$$

where $T = 24\text{h} \times 60\frac{\text{m}}{\text{h}} \times 60\frac{\text{s}}{\text{m}} = 86400\text{s}$ is one day in seconds. Set $\bar{x} = 20^\circ$, and write the dynamics in the coordinates $z := x - \bar{x}$.

- (c) Note that driving the variable $z(t) \rightarrow 0$ is therefore equivalent to driving $x(t) \rightarrow 20^\circ$. Write a `matlab` script to solve the finite-time horizon LQR problem. Select a positive semi-definite Q and a positive-definite R that gives you “decent” results. Assume the initial conditions, disturbance, and ambient temperature follow:

$$x_i(0) = 15 - i \times \text{sign}\{(-1)^i\} \quad [^\circ \text{C}] \quad (10)$$

$$d(t) = \max \left[0, 500 \sin \left(\frac{2\pi t}{86400\text{s}} \right) - 300 \right] \quad [\text{W}] \quad (11)$$

$$x_a(t) = 20^\circ + 5 \sin \left(\frac{2\pi t}{86400\text{s}} \right) \quad [^\circ \text{C}] \quad (12)$$

where the sign function is:

$$\text{sign}(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ 0 & x = 0. \end{cases} \quad (13)$$

Note: If solving the finite-time problem proves too difficult, feel free to solve the infinite-time problem instead. **Matlab** has a built-in function called **lqr** that produces the (constant) gain K ; read the documentation for details.

(d) Provide:

- A description of your simulation code
- Plots of $d(t)$, $x_a(t)$, $x_i(t)$, and $u(t)$.
- A discussion of the effects of varying Q and R

Ans:

(a) To find the dynamical equations of the building in state-space form, we have to find the system on the form

$$\dot{x} = Ax + Bu, y = Cx + Du$$

The individual room equations are

$$C_i \dot{x}_i(t) = \sum_{j \in \text{neighbors}} h_{ij} (x_j(t) - x_i(t)) + h_{i0} (x_a(t) - x_i(t)) + u_i(t) + d_i(t)$$

We rearrange this to get

$$\dot{x} = \sum_j \frac{h_{ij}}{C_i} (x_j(t) - x_i(t)) + \frac{h_{i0}}{C_i} (x_a(t) - x_i(t)) + \frac{u_i(t)}{C_i} + \frac{d_i(t)}{C_i}$$

We have 5 rooms, thus 5 states, so $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \\ x_5(t) \end{bmatrix}$, 3 controllers that

regulate temperature, $u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix}$ and finally, one disturbance term for

every room $d(t) = \begin{bmatrix} d_1(t) \\ d_2(t) \\ d_3(t) \\ d_4(t) \\ d_5(t) \end{bmatrix}$

$$\frac{h_{ij}}{C_i} = \begin{cases} 10W/^{\circ}K, & \text{if } i \text{ adjacent to } j \\ 0, & \text{else} \end{cases}$$

This makes A to be a banded matrix. Since all rooms are adjacent to room 5, the matrix A becomes

$$A = \begin{bmatrix} -\sum_j \frac{h_{1j}}{C_1} & \frac{h_{12}}{C_1} & 0 & 0 & \frac{h_{15}}{C_1} \\ \frac{h_{21}}{C_2} & -\sum_j \frac{h_{2j}}{C_2} & \frac{h_{23}}{C_2} & 0 & \frac{h_{25}}{C_2} \\ 0 & \frac{h_{32}}{C_3} & -\sum_j \frac{h_{3j}}{C_3} & \frac{h_{34}}{C_3} & \frac{h_{35}}{C_3} \\ 0 & 0 & \frac{h_{43}}{C_4} & -\sum_j \frac{h_{4j}}{C_4} & \frac{h_{45}}{C_4} \\ \frac{h_{51}}{C_5} & \frac{h_{52}}{C_5} & \frac{h_{53}}{C_5} & \frac{h_{54}}{C_5} & -\sum_j \frac{h_{5j}}{C_5} \end{bmatrix} = \begin{bmatrix} -30 & 10 & 0 & 0 & 10 \\ 10 & -40 & 10 & 0 & 10 \\ 0 & 10 & -40 & 10 & 10 \\ 0 & 0 & 10 & -30 & 10 \\ 10 & 10 & 10 & 10 & -50 \end{bmatrix}$$

Since there are three actuators in rooms 1, 2 and 4, the B matrix is written as

$$B = \begin{bmatrix} \frac{1}{C_1} & 0 & 0 \\ 0 & \frac{1}{C_2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_4} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{1.08 \times 10^6} & 0 & 0 \\ 0 & \frac{1}{1.08 \times 10^6} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{1.08 \times 10^6} \\ 0 & 0 & 0 \end{bmatrix}$$

We have the vector b that takes into account the ambient coupling of temperatures.

$$b = \begin{bmatrix} H_{10} \\ H_{20} \\ H_{30} \\ H_{40} \\ H_{50} \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix}$$

and D is a diagonal matrix with $\frac{1}{C_i} = \frac{1}{1.08 \times 10^6}$ for disturbance inputs.

If we now follow the notes on building control from the lecture slides of optimal control (lecture 4), we obtain the dynamic equations $\dot{x} = Ax(t) + Bu(t) + bx_a(t) + Dd(t)$

- (b) For this problem, we note that $z = x - \bar{x}$, then $x = z + \bar{x}$ and $\dot{x} = \dot{z}$ since $\bar{x}$ is a constant temperature of $20^{\circ}C \approx 293^{\circ}K$ since $1^{\circ}C = 1K$ (K = Kelvin) so we get the dynamics in the z coordinates as

$$\begin{aligned} \dot{z} &= A(z + \bar{x}) + Bu \\ &= Az + A\bar{x} + Bu \end{aligned}$$

$A\bar{x}$ can be seen as the forcing term towards the reference temperature in the system.

- (c) Solving the finite-time horizon problem seems to me very hard, as we don't have any initial points to work backwards from. So I just used the infinite-time horizon

5. (10 points) Suppose that $Q = \mathbf{0}$, the zero matrix, and $R = I$. Prove that

$$x_0^T S_0 x_0 = \min_{\{u_k\}_{k=0}^{T-1}} \sum_{k=0}^{T-1} \|u_k\|_2^2, \quad (14)$$

meaning that $J_0 = x_0^T S_0 x_0$ is the minimum control energy required to drive the system to zero.

Ans:

6. (10 points) *Crassidis & Junkins* Problem 3.14

Ans:

7. (10 points) *Crassidis & Junkins* Problem 3.15

Ans:

8. (10 points) *Crassidis & Junkins* Problem 7.4

Ans: